

ABDULRAHMAN ALMUTAIRI

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Objective

Electrical and Computer Engineering graduate with experience in embedded systems, PCB design, cybersecurity, and software development through hands-on engineering projects and system integration work. Seeking an engineering opportunity where I can apply my technical and problem-solving skills to develop innovative and reliable solutions.

Education

University of Colorado at Boulder

Sep. 2022 – May 2026

Bachelor of Science in Electrical and Computer Engineering

Boulder, CO

GPA: 3.410

Relevant Coursework

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|--------------------------|--------------------|-------------------------|-------------------------|
| • Senior Capstone Design | • Circuits | • RTOS | • Software Development |
| • Data Structures | • Microelectronics | • Computer Organization | • PCB Design and Layout |
| • Digital Logic/Systems | • Discrete Math | • Computer Security | |
| • Embedded Software | | | |

Projects

NASA JPL Mars Hard Lander Avionics System | STM32, Embedded Systems, RTOS, PCB Design 2025-2026

- Designed and integrated avionics subsystems for a NASA JPL-sponsored Mars hard-lander project using STM32 microcontrollers and embedded systems concepts.
- Worked on sensor integration, power distribution, data handling, and PCB design for environmental sensing and wireless transmission systems.
- Contributed to system-level hardware and firmware integration, testing, and subsystem validation.

Board 4 – Custom 4-Layer Instrument Droid PCB | Altium Designer, Embedded Hardware Apr 2025

- Designed and built a custom 4-layer Arduino-compatible PCB capable of measuring Thevenin parameters using programmable load stages and precision ADC measurements.
- Integrated ADCs, DACs, MOSFET switching circuitry, and microcontroller interfaces for embedded instrumentation applications.

Full-Stack Sports Analytics Web Platform | Node.js, PostgreSQL, JavaScript 2025

- Developed a full-stack sports analytics and prediction platform using Node.js, Express, PostgreSQL, and Handlebars in an Agile team environment.
- Implemented backend routing, authentication, database integration, and responsive frontend functionality for real-time sports data visualization and user interaction.

Machine Learning Adversarial Attack Project | Python, Neural Networks, Cybersecurity 2024

- Implemented and evaluated adversarial attacks on neural network models to analyze vulnerabilities in machine learning systems within a controlled academic environment.
- Applied concepts including neural networks, backpropagation, gradient-based attacks, and image classification to study model robustness and secure AI systems.

Technical and Professional Skills

Languages: C, C++, Python, JavaScript, SQL, HTML/CSS

Embedded Systems: STM32, Arduino, ATmega328P, RTOS, Sensor Integration

Developer Tools: STM32CubeIDE, VS Code, Docker, Git

PCB Design: Altium Designer, KiCad

Cybersecurity: Web Security, Network Security, Machine Learning Security

Core Competencies: Leadership, Team Collaboration, Problem Solving, Engineering Management, Entrepreneurship, Marketing

Operating Systems: Linux, Windows, MacOS